

Lynbrook Math Summer Camp 2026

C1 Permutations and Combinations

Lynbrook Math Club

August 2026

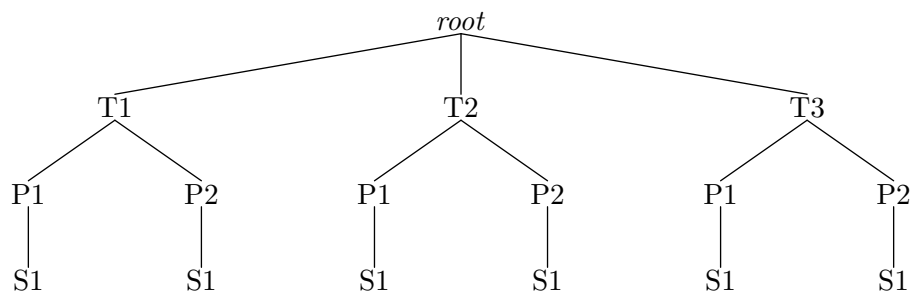
C1.1 Lecture

We make choices every day. What clothes should I wear? What order should we stand in line? In combinatorics, we learn how to count the number of possible choices quickly and elegantly without having to list them all out.

Example 1 (Multiplication Principle)

Sohum has 3 t-shirts, 2 pairs of pants, and 1 pair of shoes. How many ways can he choose one shirt, one pair of pants, and one pair of shoes to wear?

If we were to draw a decision tree for this problem, the first branch would split into 3 paths (for the shirts). From the end of *each* of those 3 paths, we would draw 2 new branches (for the pants), and from each of those, 1 branch (for the shoes).



Instead of drawing the whole tree every time, we can just multiply the number of branches at each step! Sohum has:

$$3 \text{ shirts} \times 2 \text{ pants} \times 1 \text{ shoe} = 6 \text{ outfits.}$$

This brings us to our first fundamental rule of counting.

Definition C1.1.2 (The Multiplication Principle). If a task can be broken down into a sequence of independent steps, and the first step can be done in a ways, the second step in b ways, the third in c ways, and so on, then the total number of ways to complete the task is $a \times b \times c \times \dots$

Permutations

What if we want to figure out how many ways we can arrange a specific set of items in a row?

Definition C1.1.3. A **permutation** of a set of distinct objects is a rearrangement of those objects into an ordered sequence, without repetition.

Let's start small. How many permutations of the numbers 1, 2, 3 are there? We could just list out all the possibilities:

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$$

There are 6 ways. But what if we had 10 numbers? Listing them out would take forever! Let's use the Multiplication Principle.

We have 3 spots to fill.

- For the **first** spot, we can choose any of the 3 numbers.
- For the **second** spot, one number is already used, so we only have 2 choices left.
- For the **third** spot, there is only 1 number left.

Multiplying these together, we get $3 \times 2 \times 1 = 6$ ways. This descending multiplication pattern shows up constantly in counting, so we give it a special name and symbol.

Definition C1.1.4 (Factorial). Let $n \geq 0$ be a non-negative integer. We define $n!$ (read “ n factorial”) to be the product of all positive integers between 1 and n inclusive:

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$$

By mathematical convention, we define $0! = 1$.

Claim C1.1.5 — The number of permutations (orderings) of n different objects is $n!$.

Let's do some quick factorial arithmetic. What is $\frac{9!}{7!}$? Writing it out makes it easy to simplify:

$$\frac{9!}{7!} = \frac{9 \times 8 \times 7 \times 6 \times \cdots \times 1}{7 \times 6 \times \cdots \times 1} = 9 \times 8 = 72$$

Restrictions and Choosing

Sometimes, people or objects have special conditions.

Example 6 (Group Method)

How many ways can 5 people stand in a line if Nitin and Abhigyan insist on standing right next to each other?

Since Nitin and Abhigyan must be together, let's put them in a giant sack and treat them as a single “block.” Now we are arranging 4 items: 3 individual people + 1 block. We can arrange these 4 items in $4! = 24$ ways.

However, *inside* the block, Nitin and Abhigyan can stand in $2! = 2$ ways (Nitin first, or Abhigyan first). By the Multiplication Principle, the total number of ways is $4! \times 2! = 24 \times 2 = 48$ ways.

What if we only want to arrange *some* of the objects, not all of them?

Example 7

A town council has 8 members. How many ways can they elect a President, a Vice-President, and a Secretary?

We have 3 specific positions to fill.

- President: 8 choices.
- Vice-President: 7 choices remaining.
- Secretary: 6 choices remaining.

By the Multiplication Principle, there are $8 \times 7 \times 6 = 336$ ways.

This leads us to our generalized formula for choosing and arranging k objects out of n .

Definition C1.1.8 ($P(n, k)$). Let n and k be non-negative integers with $k \leq n$. The number of ways to choose k objects from a collection of n distinct objects, where the **ordering** of the k objects matters, is denoted $P(n, k)$ and is equal to:

$$P(n, k) = \frac{n!}{(n-k)!} = n \times (n-1) \times \cdots \times (n-k+1)$$

Combinations

We define the word **combination** as used in combinatorics as follows:

Definition C1.1.9. A **combination** (or k -combination) is a way of selecting k objects from a collection of n different objects, where the order of the objects selected does not matter.

Example 10

The same town council has 8 members. This time, they just want to form a 3-person committee. How many ways can this committee be formed?

Before we solve this, ask yourself: what is the difference between this example and Example 7? In Example 7, we were choosing three members to serve three *different* roles. Here, we are only selecting three members without assigning particular roles to each member.

As before, there are $P(8, 3) = 8 \times 7 \times 6$ ways to select a first, second, and third member. However, this overcounts by a factor of $3! = 6$: we have counted the 3-person committee {Alice, Bob, Carl} six times (e.g., select Alice first, Bob second, Carl third; select Bob first, Carl second, Alice third, etc.).

Because each 3-person committee must be counted once and not 6 times, we divide by 6 to correct for our overcounting. The number of ways to select three members for a committee is

$$\frac{P(8, 3)}{3!} = \frac{8 \times 7 \times 6}{6} = 8 \times 7 = 56.$$

Definition C1.1.11 (Binomial Coefficient). Let n and k be integers with $0 \leq k \leq n$. The notation $\binom{n}{k}$ (read as “ n choose k ”) represents the number of k -object combinations which can be selected from a collection of n different objects.

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{(n-k)!k!} = \frac{n(n-1)(n-2) \cdots (n-k+1)}{k!}$$

Let's do some quick calculation exercises to get comfortable with the formula:

- $\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10$
- $\binom{4}{1} = \frac{4}{1} = 4$
- $\binom{5}{3} = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = \frac{60}{6} = 10$

Wait, notice anything interesting? $\binom{5}{2}$ and $\binom{5}{3}$ both equal 10. This isn't a coincidence!

Claim C1.1.12 (Symmetry of Combinations) — Let n and k be integers with $0 \leq k \leq n$. Then:

$$\binom{n}{k} = \binom{n}{n-k}$$

The idea is to show that the binomial coefficients $\binom{n}{k}$ and $\binom{n}{n-k}$ count the same thing, and therefore must be equal to each other. Here, $\binom{n}{k}$ equals the number of ways to select k objects from a collection of n different objects. However, we can equivalently select k objects (to form a group, committee, etc.) by selecting $n - k$ objects *not* in that group or committee, and the number of ways to do this is $\binom{n}{n-k}$. Therefore, $\binom{n}{k} = \binom{n}{n-k}$.